

Digi-Chalk Business Model

₹23,18,850 | TRL2→TRL9

Digi-Chalk – Team



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Problem Statement

The Problem

- Smart-boards exist, but they're priced for schools that can already afford them – most Indian government and low-income schools are locked out entirely
- Blackboard teaching leaves no digital trace – lessons can't be saved, shared, or revisited by students or teachers
- Existing “digitization” solutions require replacing the blackboard entirely, an unrealistic cost for under-resourced schools
- Result: millions of classrooms have no path to digital learning infrastructure, not because the content isn't valuable, but because the hardware cost is prohibitive



Erased = Gone Forever.

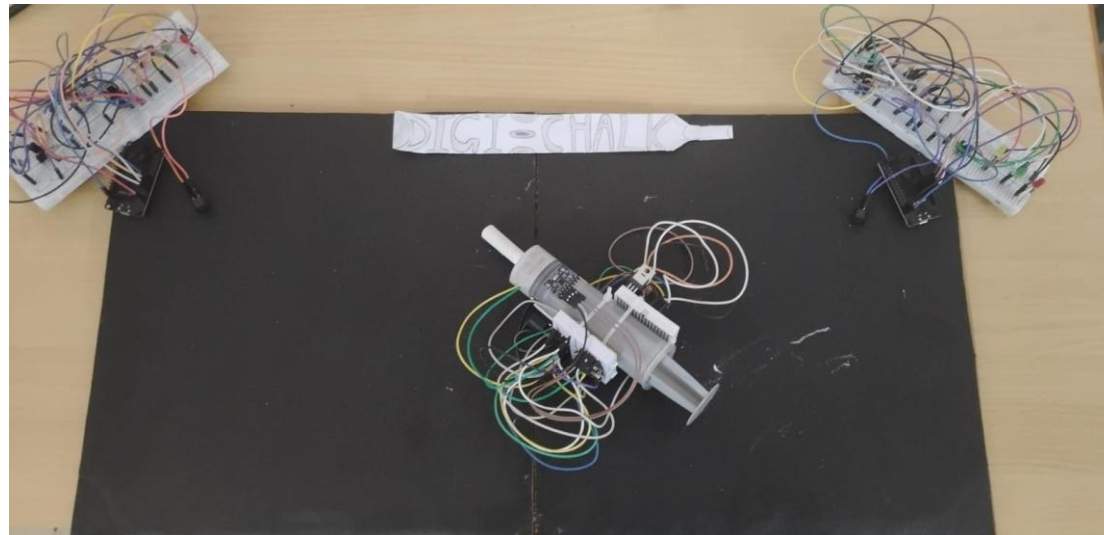


Every Stroke, Every Word, Saved.

Solution – Product Brief

Digi-Chalk: Edge-Computed Chalk Holder Retrofit for Digitization of Existing Blackboards

- A retrofit hardware module — no need to replace the blackboard, just add Digi-Chalk to it
- Core sensing: IMU motion tracking (MPU6050) + ultrasonic TDOA triangulation (MA40S4S/R, ICS-43434) + Decawave DWM1000 UWB for robust wireless communication link
- XIAO ESP32-C3 based compute across both the Chalk Holder and Corner Clips, for a compact, consistent architecture
- Rechargeable NiMH power system with dedicated boost regulation — no disposable batteries, no rural sourcing dependency
- Captures and digitizes lesson content as it's written, without disrupting how teachers already teach
- Provisional patent filed (Application No. 202641089516)



What's in the Kit

Kit Component	Qty	What It Does
Chalk Holder Stylus	1	Wireless, rechargeable, tracks handwriting in real time
Corner Clips (receivers)	2	Mount to blackboard corners, capture position data
5,000mAh Power Banks	4 (2 active + 2 spare)	Power the clips, swappable without downtime
NiMH Rechargeable AAA Set	1 set	Powers the chalk holder, 2,100+ recharge cycles
Neodymium Magnets + VHB Steel Anchor Discs	2	Permanently adhere to the blackboard frame; clips snap on/off magnetically for easy mounting and removal
Calibration Alignment Strip	1	One-time setup for precise tracking
Charging Briefcase/Dock	1	Recharges everything overnight, portable
1-Year SaaS License	1	Cloud backup, AI tutoring, stroke smoothing

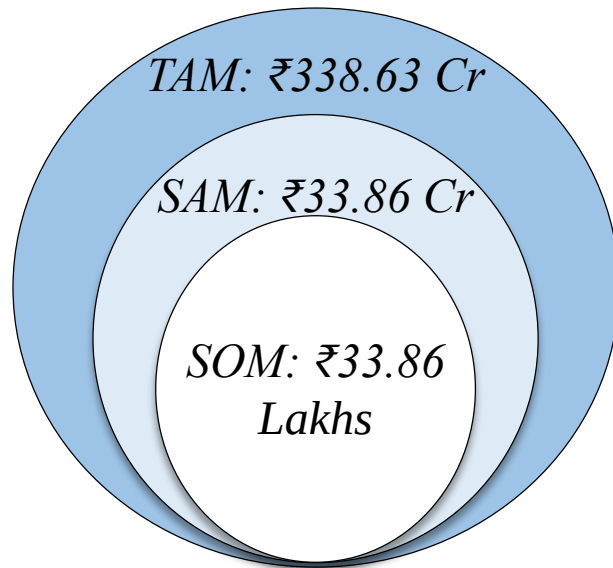


Unique Selling Points / Impact

Why Digi-Chalk Wins

- **Retrofit, not replacement** – a fraction of the cost of a full smart-board install
- **Built for the schools smart-board vendors ignore:** government and low-income schools, not premium private schools
- **Key persona:** District/ Block Education Officer or Headmaster at a Tamil Nadu government school – the actual decision-maker for a classroom infrastructure purchases under state education budgets (e.g. Samagra Shiksha / SSA funding routes), not a private buyer. **Secondary persona:** State School Education Department procurement official, for a scaled/bulk deployment.

Assumption	Value
Government schools in Tamil Nadu	37,626 (as of 2024-25, per Union Government data)
Avg. units per school (1 per classroom, pilot config)	2 units
Price per unit	₹45,000
Realistic serviceable share (Year 1-2, pilot + early scale)	1% of schools



- **TAM** (Tamil Nadu, full penetration): $37,626 \text{ schools} * 2 \text{ units} * ₹45,000 = ₹338.63 \text{ Cr}$
- **SAM** (3-year target, 10% of schools): $3,763 * 2 * ₹45,000 = ₹33.86 \text{ Cr}$
- **SOM** (Year 1-2 pilot-to-early-scale, 1% of schools): $376 * 2 * ₹45,000 = ₹33.86 \text{ L}$

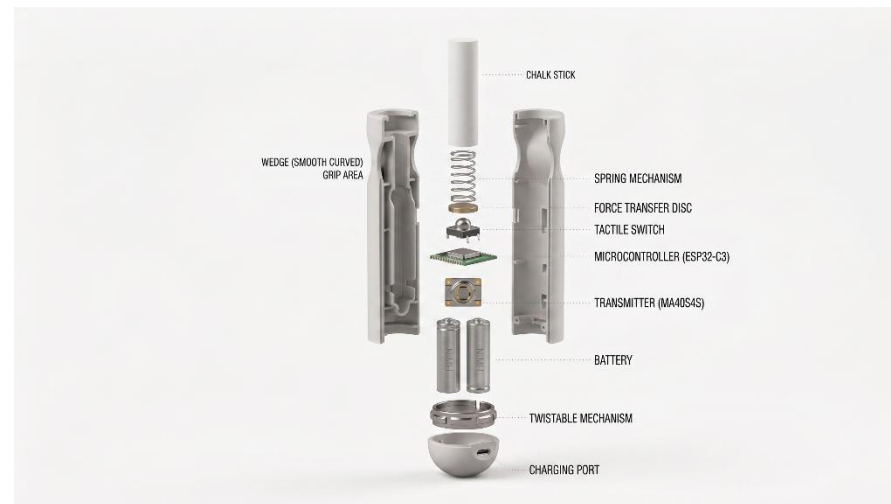
Competitive Analysis

Player	Price Point	Target Segment	Approach
Premium Interactive Panels (Samsung, LG, ViewSonic, BenQ)	₹1,50,000 – ₹4,50,000+	Private/premium schools, higher-ed lecture halls	Full board replacement, wall-mounted display
Budget/Value Interactive Panels (Evota, iSLate, iBall, Maxhub)	₹55,000 – ₹1,80,000	Mid-tier private, some govt schools	Full board replacement, lower-cost entry tier
Hybrid Chalk-Digital Boards (Chinese manufacturers, e.g., Nano Smart Blackboard)	Not publicly listed — likely premium, custom-quoted	Institutions wanting to retain chalk with embedded digital sync	Full board replacement — chalk-writable surface + embedded display panel built into one new board
Digi-Chalk	₹45,000	Government/low-income schools	Retrofit onto existing blackboard, no replacement

- Digi-Chalk is priced 45–75% below the interactive display panel market, depending on tier
- Even the closest conceptual competitor — hybrid chalk-and-digital boards — still requires buying an entirely new board; Digi-Chalk adds digitization to the blackboard a school already owns, with zero furniture replacement and zero rewiring
- We're creating a new mid-cost category — priced above disappearing budget projector kits, but meaningfully below every named interactive panel brand — accessible to schools operating under SSA/state grant budgets

Revenue Model & Unit Economics

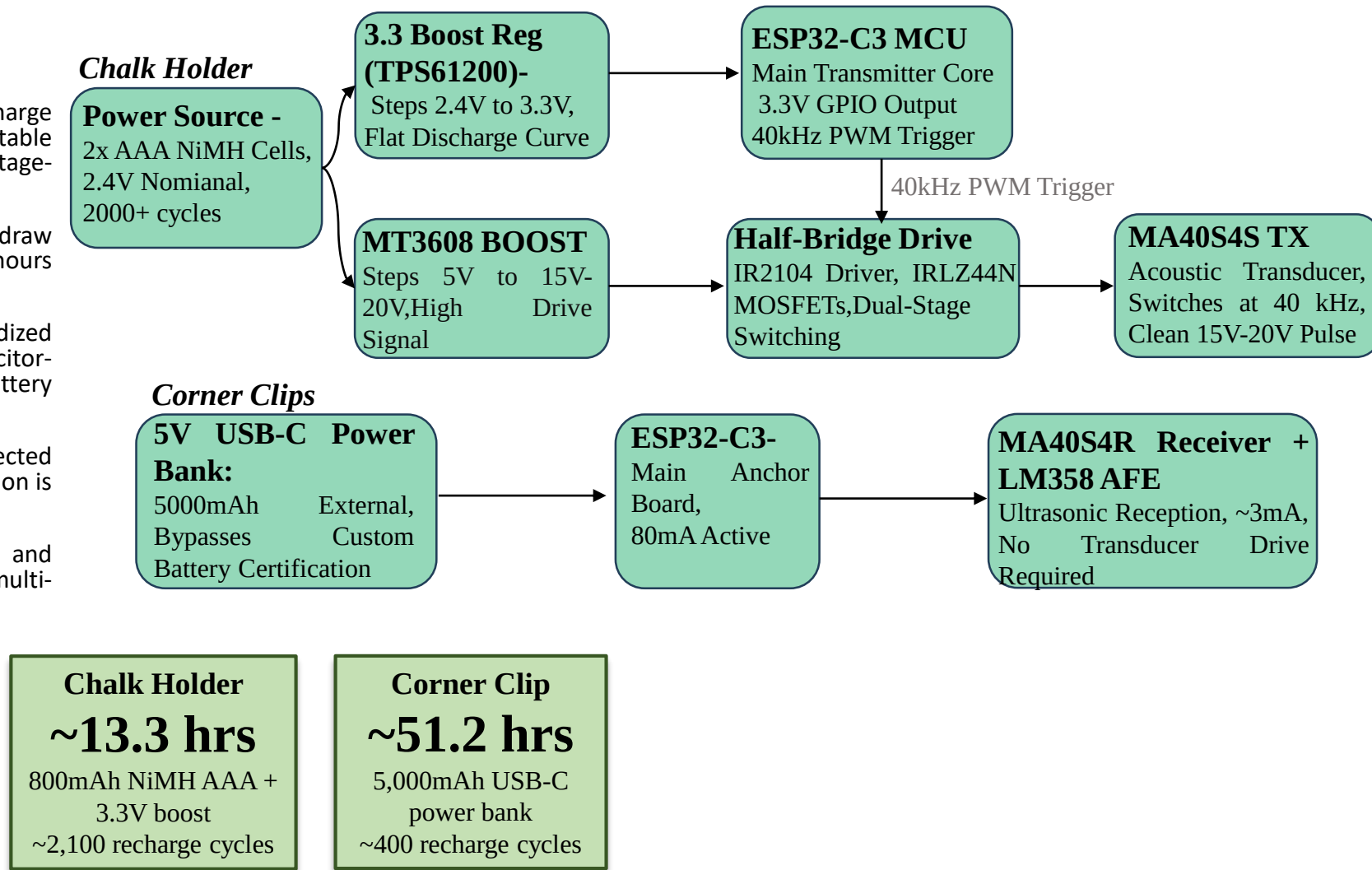
- **Revenue Model:** Hardware one-time sale + recurring SaaS subscription (B2B Hardware + SaaS hybrid)
- **Hardware Pricing:** ₹45,000 per kit (2-clip system + chalk holder)
- **Unit Cost (COGS):** ₹12,790 at 1,000-unit volume production
- **Hardware Gross Margin:** 71.6% (3.52x COGS multiple)
- **SaaS Layer:** bundled into kit price for Year 1; renews at ₹2,000/month per classroom from Year 2 — cloud backups, RAG-based tutoring, PAMO stroke smoothing (95%+ margin)
- **3-Year Blended Gross Margin:** ~86%
- **Sales Channels:** Direct/private school sales (shorter cycle) + Government tender/procurement via Samagra Shiksha/SSA (longer cycle, bulk volume)



Power Architecture

Rural-Ready, De-Risked Power System

- **Chalk Holder:** 2× NiMH AAA rechargeable cells (2,100+ recharge cycles) + dedicated 3.3V boost regulator, guaranteeing stable voltage across the full discharge curve — eliminates voltage-sag/brownout risk of disposable alkaline
- **Wake-on-Write (WOW):** hardware interrupt drops standby draw to 2.0 mA; blended average draw of 50.92 mA yields ~13.3 hours runtime per charge
- **Corner Clips:** always-on receivers, powered by standardized 5,000 mAh power banks (35.4 hrs/charge), with supercapacitor-bridged hot-swap for uninterrupted tracking during battery changes
- Supercapacitor-bridged hot-swap: system remains connected and calibrated during a battery change — exact bridge duration is being finalized against final hardware validation.
- No custom lithium-battery certification burden — NiMH and externally-certified power banks avoid the additional multi-month lithium-specific regulatory pathway



Business Roadmap: TRL 2 to TRL 9

Phase 0 — TRL2 → TRL3 (Founder-Funded, In Progress)

- Breadboard proof-of-concept: validate IMU + ultrasonic TDOA + UWB fusion together in real time
- Fabricate and assemble first 3-unit PCB batch, flash firmware
- Full bench validation — confirm sensing pipeline on final hardware

Phase A — TRL3 → TRL6 (Months 1–6)

- Select PCB fabrication partner (local Coimbatore vs. JLCPCB/PCBWay)
- Produce 10-15 unit small batch
- Classroom-condition testing: durability, power stability, non-technical user handling
- Begin StartupTN pilot partner conversations
- Budget: ₹2,64,850 (Investor Ask)

Phase B — TRL6 → TRL9 (Months 7–15)

- Scale to 100+ unit production run with locked manufacturing partner
- Multi-district pilot deployment (3-5 districts) across a full academic term
- BIS/CRS + WPC wireless certification for government tender eligibility
- Company incorporation, GST setup, import duty clearance
- Teacher training rollout at scale across pilot sites
- Field support/ops layer across the team
- Budget: ₹20,54,000 (Investor Ask)



Phase wise breakdown

Phase A Breakdown (₹2,64,850) — Investor Ask

Item	Amount (₹)
Component + fabrication + assembly (12 units avg)	1,97,850
Field testing logistics	28,000
Certification/compliance checks	22,000
Iteration buffer	17,000
Total	2,64,850

Phase B Breakdown (₹20,54,000) — Investor Ask

Item	Amount (₹)
100-unit production run (true volume pricing, ₹12,790/unit)	12,79,000
Multi-district pilot deployment (3-5 districts)	2,50,000
BIS/CRS + WPC wireless certification	1,15,000
Company incorporation + GST setup	15,000
Import duty buffer	20,000
Teacher training rollout at scale	80,000
Field support/ops (stipend, travel across pilot term)	1,20,000
Contingency	1,00,000
Iteration/pilot-revenue reconciliation buffer	75,000
Total	20,54,000

Compliance & Regulatory Costs

Item	Estimated Cost	Purpose
BIS/CRS Certification (Electronics Safety)	₹1,00,000	Government tender eligibility
WPC/ETA Wireless Equipment Approval	₹15,000 – ₹30,000	Type approval for UWB/wireless RF components
Company Incorporation (Private Limited)	₹10,000 – ₹20,000	Government + professional filing fees
GST Registration & Compliance Setup	₹5,000 – ₹10,000	Registration + ongoing filing
Import Duty Buffer (DWM1000, MA40S4R, sensors)	₹15,000 – ₹25,000	Customs duty on imported components at volume

Included within the TRL9 budget above — reflects full regulatory readiness, not just BIS certification.

Budget Summary

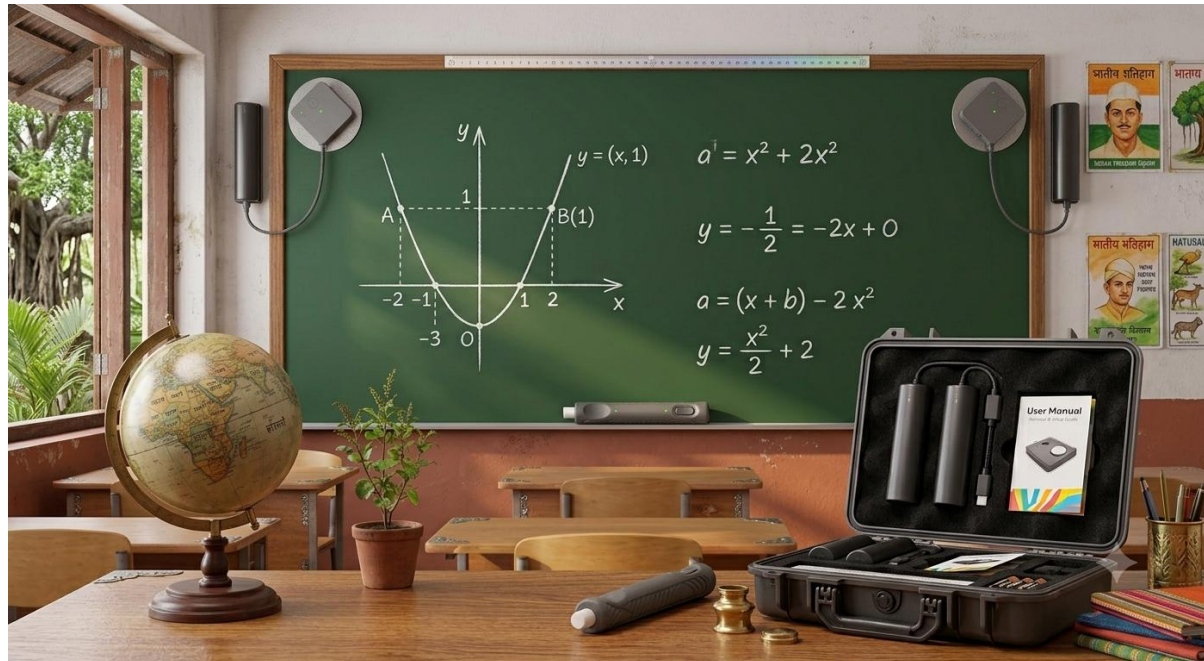
TRL Milestone	Budget (₹)	Funded By	Key Deliverable
TRL2 → TRL3	1,23,400 (Approx.)	Founder (in progress)	Bench-validated breadboard prototype
TRL3 → TRL6	2,64,850	Investor Ask	Small-batch field-tested prototype
TRL6 → TRL9	20,54,000	Investor Ask	Multi-district pilot, certified, production-scaled
Total Investor Ask	23,18,850		—

Risk Mitigation

Risk	Mitigation Strategy
No fab partner locked yet	Front-loaded decision in Month 1
First-time PCB fabrication risk	Small-batch-first approach (3 units, then 10-15)
Early TRL (2) — core sensing not yet validated	Breadboard proof-of-concept front-loaded in Phase A
Voltage sag/brownout under peak load	Resolved via NiMH + dedicated 3.3V boost regulator
Rural battery availability/cost	Rechargeable NiMH (2,100+ cycles) removes disposable dependency
Field-testing dependency	StartupTN engagement addresses pilot-partner gap
TRL9 field durability risk at scale	Phased multi-district rollout, not single-shot launch

Why Fund This Now

- **Design is de-risked:** provisional patent filed
- **Power architecture is de-risked:** NiMH + boost regulation solves voltage-sag and rural-availability problems directly
- **Clear, gated execution path:** TRL3 self-funded, TRL6 and TRL9 fully itemized including compliance
- Premium positioning (₹45,000, 71.6% margin) with a bundled-then-recurring SaaS layer — a fundamentally stronger unit economics story than the original ₹2,000 budget model
- ₹23,18,850 to carry Digi-Chalk from TRL2 through certified, field-proven, multi-district TRL9 deployment



Thank You

Digi-Chalk – Never Erased Again

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Digi-Chalk isn't a smarter blackboard for schools that can already afford one. It's the first affordable way to bring lesson digitization to the schools that were never going to get a smart-board at all.